

BOA JANG

Seoul National University, Interdisciplinary Program in Bioengineering

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BIOGRAPHY	I develop clinically grounded AI methods for medical imaging, centered on ophthalmic imaging and extending to digital pathology, and neuroimaging. My work spans representation learning, multi-modal prediction, image restoration, and test-time adaptation—validated against practicing clinicians to build AI systems reliable enough for real-world care.	
EDUCATION	Seoul National University	Seoul, Korea
	<i>Ph.D. Candidate in Bioengineering</i>	Mar. 2023 – Present
	Peking University	Beijing, China
	<i>B.S. in Psychological and Cognitive Sciences</i>	Sep. 2016 – Jul. 2020
RESEARCH EXPERIENCE	Seoul National University Hospital (SNUH)	Seoul, Korea
	<i>Medical AI Researcher</i>	Nov. 2021 – Feb. 2023
	Korea Institute of Science and Technology (KIST)	Seoul, Korea
	<i>Research Intern, Brain Science Institute</i>	Aug. 2020 – May 2021
	Peking University	Beijing, China
	<i>Undergraduate Research Assistant</i>	Mar. 2019 – Jun. 2020
JOURNAL PUBLICATIONS	[j.8] J.R. Song [†] , B. Jang [†] , et al. (2026). Segmentation-Guided Denoising Diffusion Probabilistic Models for Retinal OCT Speckle Noise Reduction with Enhanced Clinical Diagnostic Accuracy. <i>Scientific Reports</i> (Accepted).	
	[j.7] B. Jang , Y. Ahn, et al. (2026). Clinically Grounded Retinal Representation Learning from Minimal Supervision. <i>Scientific Reports</i> .	
	[j.6] B. Jang , Y. Shin, G. Lee, and Y.-G. Kim (2026). From Static Diagnosis to Dynamic Guidance: Evolution of Artificial Intelligence in Pediatric Neuroimaging. <i>Journal of Korean Neurosurgical Society</i> .	
	[j.5] J.H. Lee, S.Y. Lee, B. Jang , et al. (2026). OCT Biomarkers as Predictors of Treatment Interval in Neovascular Age-Related Macular Degeneration Treated with Intravitreal Aflibercept Using a Treat-and-Extend Regimen. <i>Scientific Reports</i> .	
	[j.4] B. Jang [†] , C.H. Lee [†] , et al. (2026). Artificial Intelligence-Based Prediction of First Recurrence in Neovascular Age-Related Macular Degeneration with Validation by 19 Experts. <i>Scientific Reports</i> .	
	[j.3] D.Y. Kim, R. Do, ..., B. Jang , et al. (2025). Automated AI-Based Identification of Autism Spectrum Disorder from Home Videos. <i>npj Digital Medicine</i> .	
	[j.2] N.Y. Kim [†] , B. Jang [†] , et al. (2023). Differential Diagnosis of Pleural Effusion Using Machine Learning. <i>Annals of the American Thoracic Society</i> .	
	[j.1] B. Jang [†] , S.Y. Lee [†] , et al. (2023). Preliminary Analysis of Predicting the First Recurrence in Patients with Neovascular Age-Related Macular Degeneration Using Deep Learning. <i>BMC Ophthalmology</i> .	
	[†] These authors contributed equally.	
PRESENTATIONS	[o.7] B. Jang , G. Lee, J. Choi, and Y.-G. Kim. “Structure-Preserving Test-Time Adaptation for Cross-Modality Retinal Vessel Segmentation.” <i>2026 Spring Conference of The Korean Society of Medical Informatics (KoSMI)</i> , Cheongju, Korea.	
	[po.3] B. Jang , K.H. Bae, and Y.-G. Kim. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” <i>2025 Annual Conference of the Korea Society of Artificial Intelligence in Medicine (KoSAIM)</i> , Seoul, Korea.	
	[o.6] B. Jang et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” <i>MICCAI 2025 AMAI Workshop</i> , Daejeon, Korea. Best Abstract Award – Honorable Mention.	

UNDER REVIEW	[o.5] B. Jang et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” <i>2025 Spring Conference of The Korean Society of Medical Informatics (KoSMI)</i> , Busan, Korea. Best Paper Award .
	[o.4] B. Jang et al. “Self-Supervised Learning for Mesangial Proliferation Prediction in Digital Pathology.” <i>The 45th Annual Meeting of the Korean Society of Nephrology (KSN 2025)</i> , Seoul, Korea.
	[o.3] B. Jang and Y.-G. Kim. “Foundation Models in Healthcare: Development, Necessity and Implementation Strategies.” <i>2025 HIMSS Global Health Conference & Exhibition (HIMSS 2025)</i> , Las Vegas, NV, USA.
	[o.2] B. Jang et al. “The Impact of AI-Assistance on Predicting Recurrence in Neovascular Age-Related Macular Degeneration for Ophthalmologists.” <i>2023 Fall Conference of The Korean Society of Medical Informatics (KoSMI)</i> , Bundang, Korea. Best Paper Award .
	[po.2] B. Jang et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” <i>10th Workshop on Ophthalmic Medical Image Analysis (OMIA-X), 2023 in International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)</i> , Vancouver, Canada.
	[po.1] B. Jang , Y.-G. Kim, and E.K. Lee. “Predicting the First Recurrence in Patients with Neovascular AMD Using Deep Learning.” <i>2023 American Society of Retina Specialists (ASRS)</i> , Seattle, WA, USA. Poster Award Winner .
	[o.1] B. Jang et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” <i>2023 Spring Conference of The Korean Society of Medical Informatics (KoSMI)</i> , Daegu, Korea. Best Paper Award .
	[m.5] Skeleton-Guided Progressive Test-Time Adaptation for Thin Curvilinear Structures. <i>AAAI2027</i> (under review).
	[m.4] Retinal Age Gap as a Biomarker of Biological Aging Reflected by Socioeconomic Status and Lifestyle Behaviors. <i>GeroScience</i> (in revision).
	[m.3] Prediction of Postoperative Intracerebral Hemorrhage Following Brain Biopsy Using Integrated Statistical and Machine Learning Analyses. <i>Scientific Reports</i> (in revision).
AWARDS	[m.2] Integrating RNFL and Fundus Photographs for Differentiation of Glaucomatous and Non-Glaucomatous Optic Neuropathy Using Deep Learning. <i>Scientific Reports</i> (in revision).
	[m.1] Age- and Sex-Specific Autism Screening Using Machine Learning Reduction and Conversational Agent Integration. <i>PLOS Digital Health</i> (submitted).
	[a.5] AI Seoul Tech Graduate Scholarship, Seoul Future Talent Foundation Nov. 2025
	[a.4] Hana Bank President’s Award, 2024 Digital Wellness Competition Nov. 2024
	[a.3] 2nd Place, Healthcare AI Learning Platform Challenge 2023, Themes 4 & 6 Dec. 2023
	[a.2] 3rd Place, Digital Pathology AI Hackathon, CODiPAI Challenge Oct. 2023
MENTORING	[a.1] SNUAIMED Excellence Scholarship 2023, 2024
	[t.4] Teaching Assistant, Korea Clinical Datathon (KCD) 2025 Oct. 2025
	[t.3] Student Ambassador, MICCAI 2025 AMAI Workshop Sep. 2025
	[t.2] Teaching Assistant, Clinical and Healthcare AI Training Programs 2024, 2025
	[t.1] Teaching Assistant, NVIDIA MONAI Bootcamp, HCLS Summit Korea 2022 May 2022
PATENTS	[p.2] Y.-G. Kim, . . . , B. Jang . Controllable Normal-to-Disease Fundus Translation via Factorial Guidance. Korea Patent Application No. 10-2026-0119988, filed June 2026. (<i>Pending</i>)
	[p.1] H.-S. Kim, . . . , Y.-G. Kim, B. Jang . Method and System for Predicting Information Related to Abdominal Surgery from Medical Image Data. Korea Patent No. 10-2953173, registered April 2026. (<i>Granted</i>)
REFERENCES	Prof. Jin-Wook Choi, PhD (jinchoi@snu.ac.kr) Prof. Young-Gon Kim, PhD (younggon2.kim@gmail.com) Prof. Hyuk Jin Choi, MD, PhD (docchoi@hanmail.net)